Dual-LED Flasher Circuit

This project demonstrates how to build a simple 555 timer circuit with two flashing LEDs, both flashing at the same rate

Parts List

Required Parts

Number Required	Part Name	Notes
1	9-Volt Battery	
1	9-Volt Battery Wiring Harness	Amazon
1	Breadboard	Amazon
4	* Jumper Wires	Amazon
1	555 Timer Chip	Amazon
1	1uF Electrolytic Capacitor	Amazon
1	** 100K Ohm Resistor	Amazon
1	** 470K Ohm Resistor	Amazon
2	** 270 Ohm Resistor	Amazon
2	Light-Emitting Diode (LED)	Amazon

* Four short jumper wires, each 2-inches or less

** Standard non-polarized resistor. That is, it dosen't matter which direction current flows through the resistor, so it won't matter how you connect it

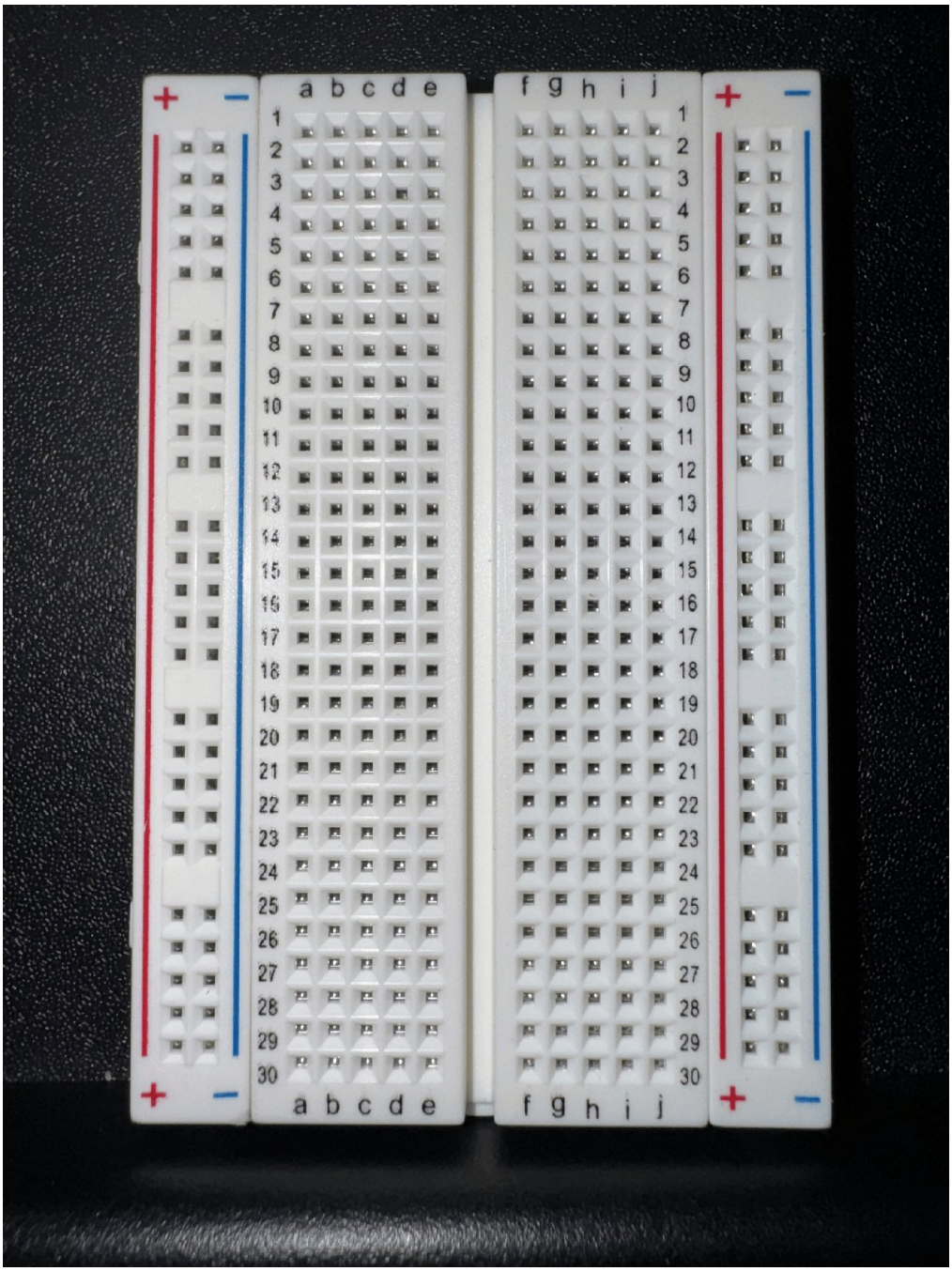
The capacitor and LEDs *are* polarized, so you'll need to identify their positive (+) and negative (-) terminal wires to ensure that current flows through them in the correct direction. The build steps below will explain how to identify and properly connect them in the circuit.

Optional Parts

Part Name	Notes
Multi-function Wire Stripping/Cutting Pliers	For trimming and stripping wires as needed (Amazon)

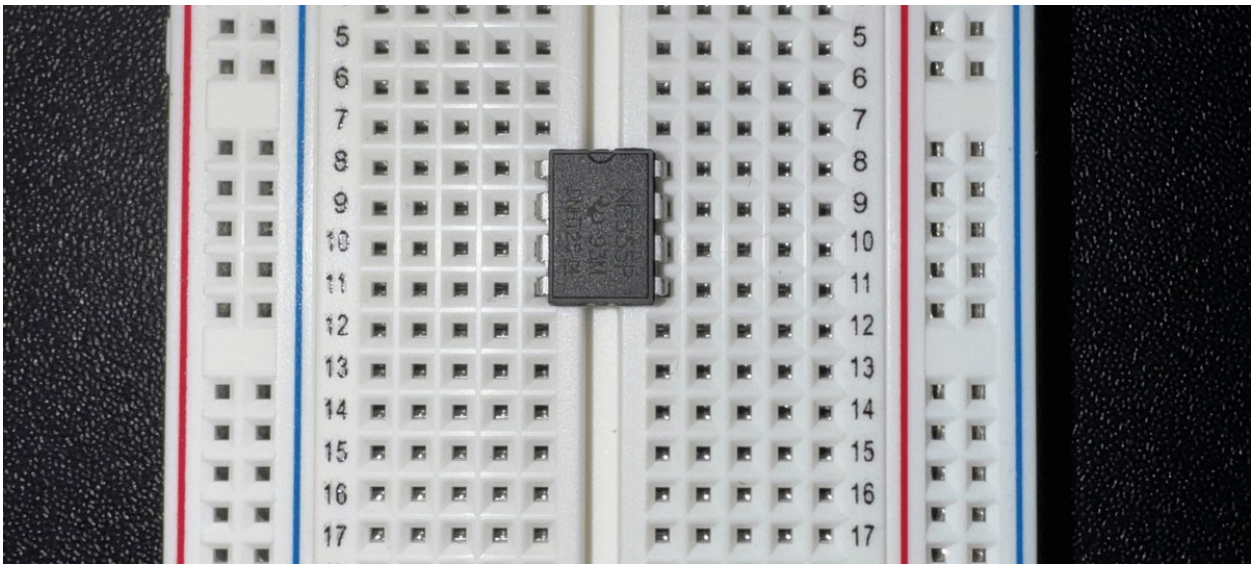
Step 1: Breadboard Orientation

Place the breadboard on a flat surface, and orient it so that Row 1 is at the top from your point of view



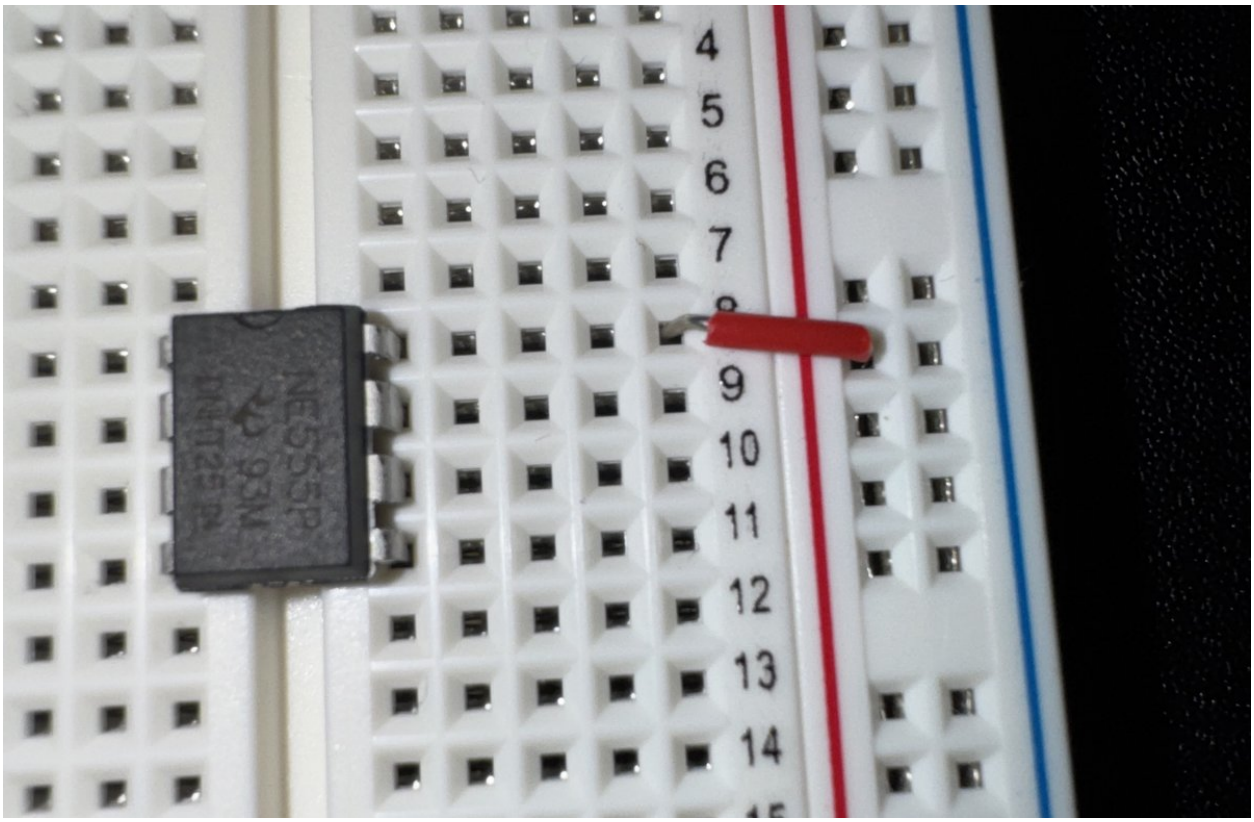
Step 2: Add the 555 Chip

- Carefully insert your 555 timer chip with its half-moon notch pointing upward, toward the top of the breadboard
- The 4 pins on each side of the 555 should span the wide center channel that vertically divides the left and right halves of your breadboard
- The horizontal row that you choose for inserting the 555 is up to you, but make sure to leave plenty of room below the 555 to expand downward with the remaining components
- Anywhere between row 5 and row 11 should be fine



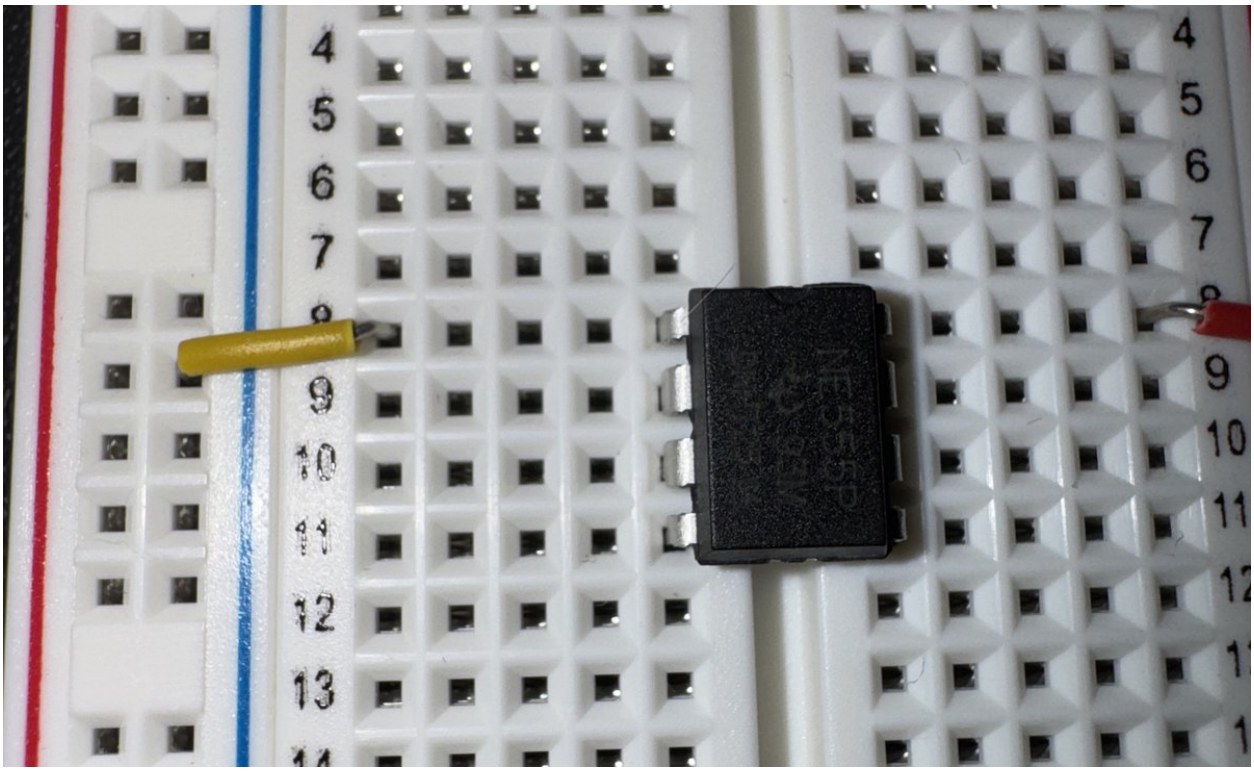
Step 3: Connect 555's VCC [8] to Breadboard's Power Rail (+)

Use a short jumper wire to connect the 555's **VCC** pin **[8]** to the breadboard's red power rail **(+)**



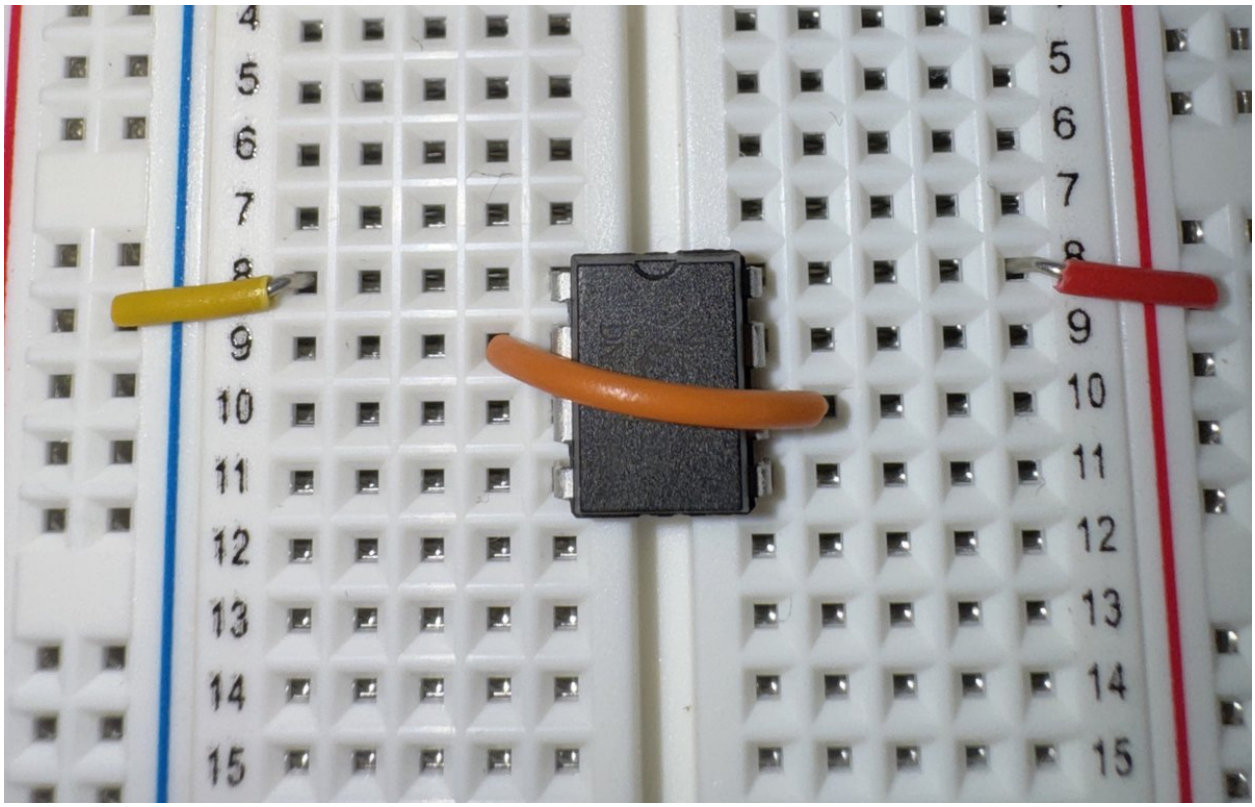
Step 4: Connect 555's GND [1] to Breadboard's Ground Rail (-)

Use a short jumper wire to connect the 555's **GND** pin **[1]** to the breadboard's blue ground rail **(-)**



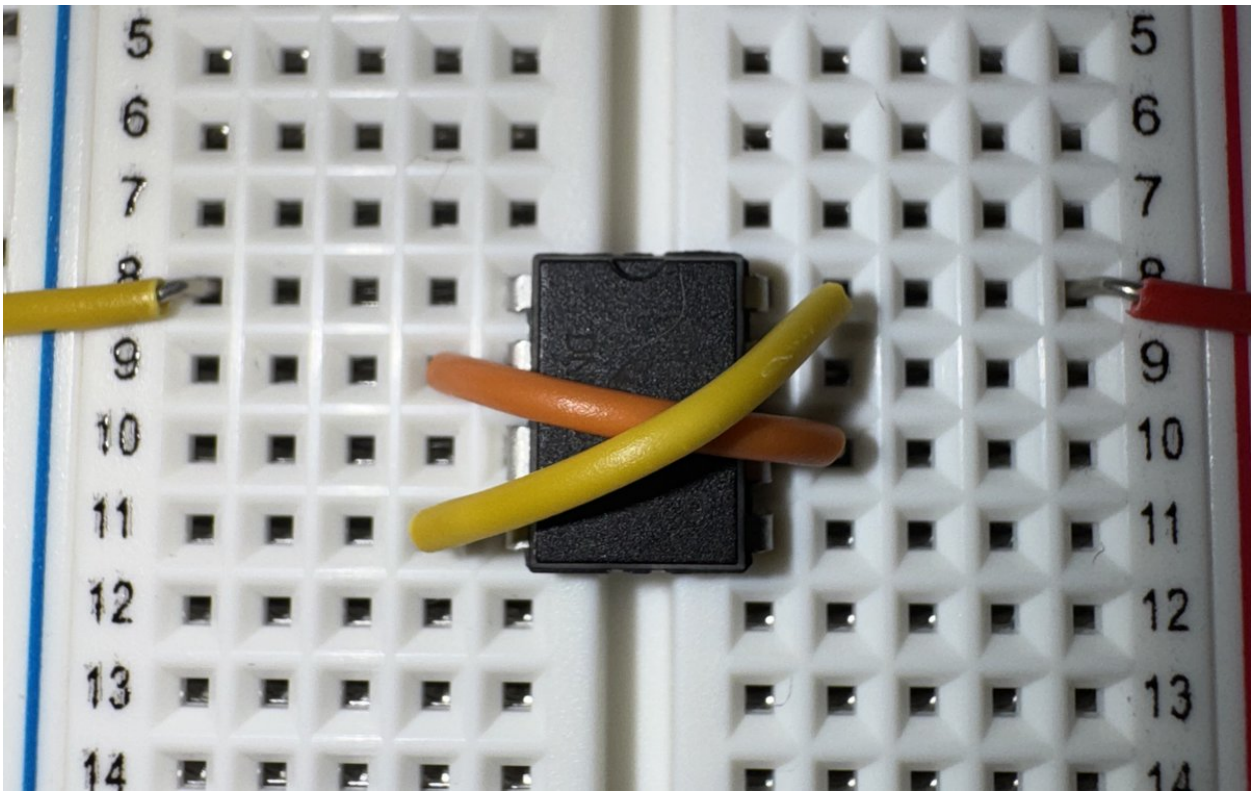
Step 5: Connect 555's TRIGGER [2] and THRESHOLD [6] pins

Use a medium jumper wire to connect the 555's **TRIGGER** pin **[2]** and **THRESHOLD** pin **[6]**



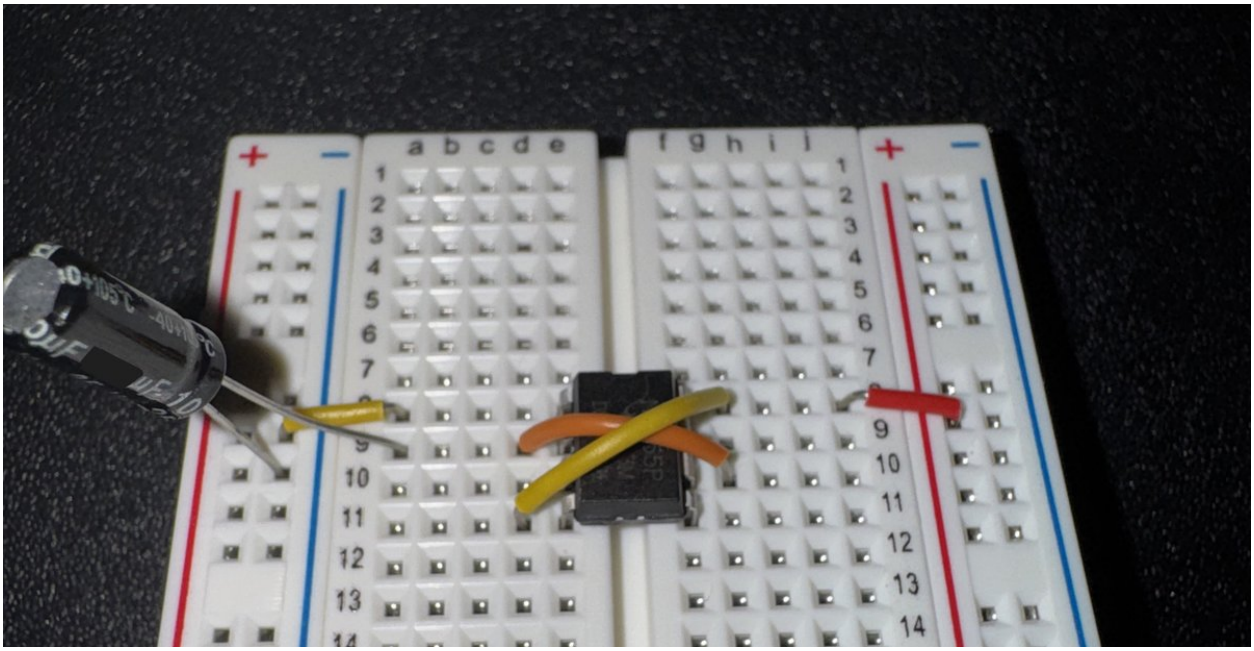
Step 6: Connect 555's RESET [4] and VCC [8] pins

Use another medium jumper wire to connect the 555's **RESET** pin **[4]** and **VCC** pin **[8]**



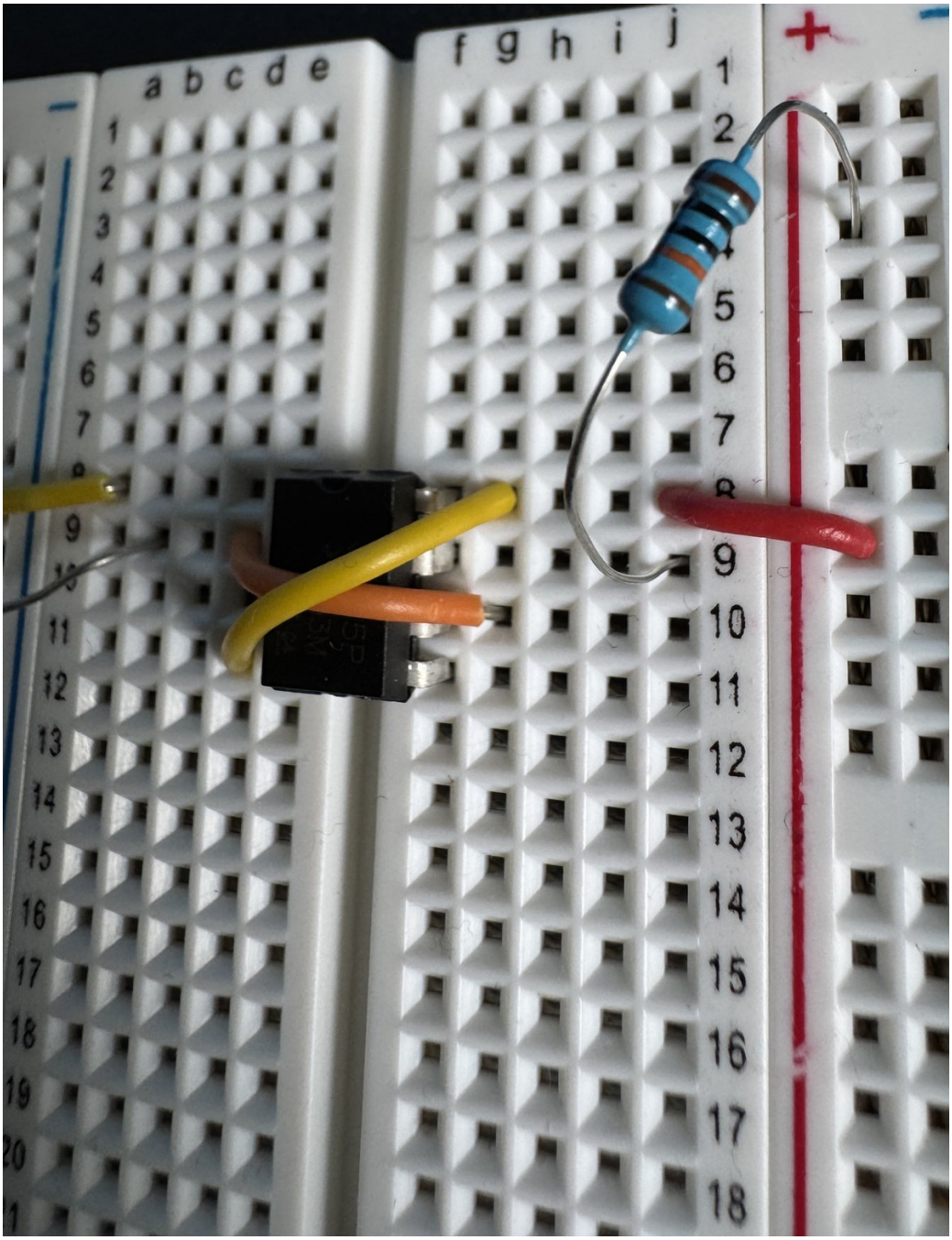
Step 7: Connect 1uF Capacitor to 555's TRIGGER [2] and Ground Rail (-)

- Connect the capacitor's positive pole (long leg, "anode") to the 555's **TRIGGER** pin **[2]**
- Connect the capacitor's negative pole (short leg, "cathode") to the left side ground rail **(-)**



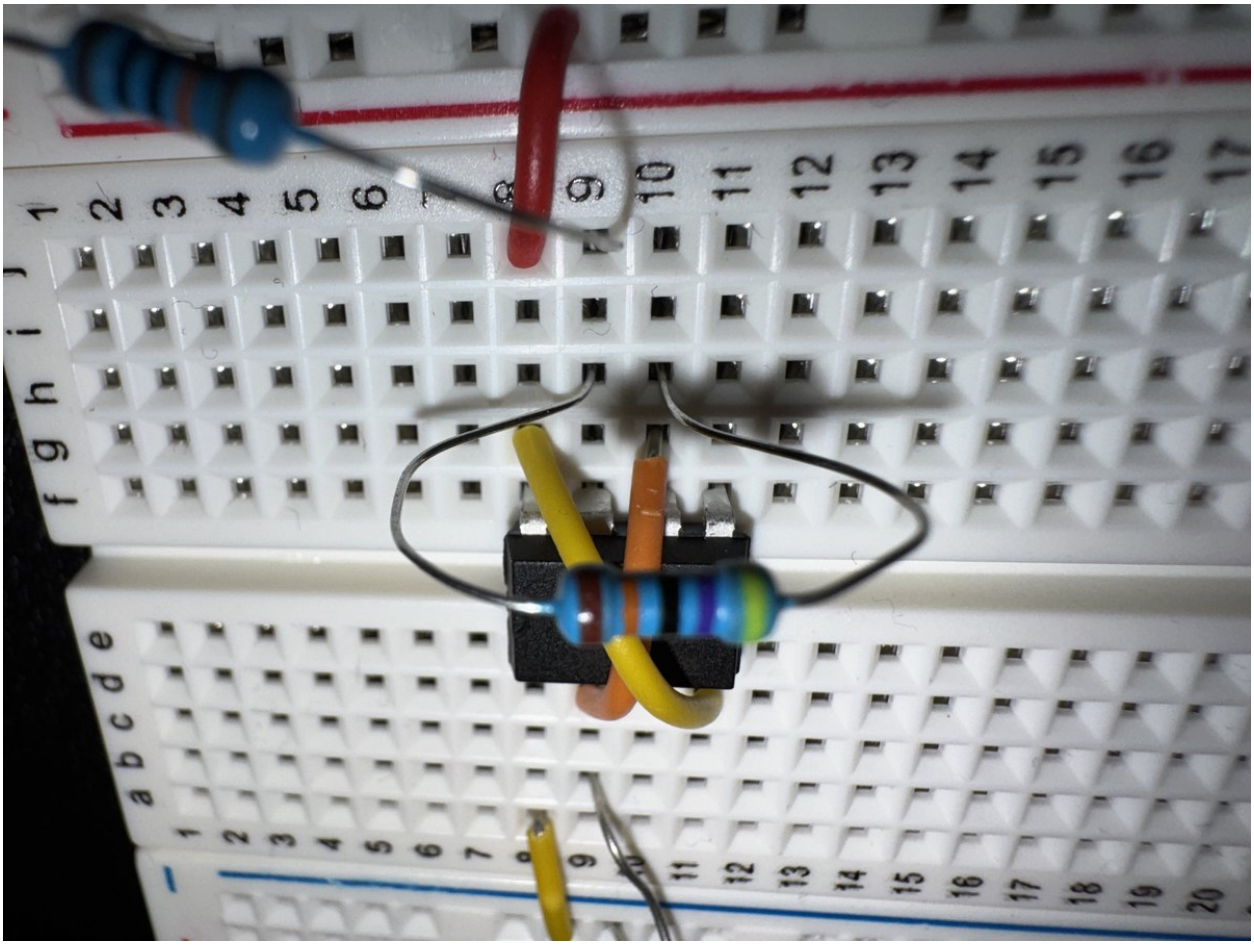
Step 8: Connect 100K Ohm Resistor to 555's DISCHARGE [7] and Power Rail (+)

- Connect one end of the 100K ohm resistor to the 555's **DISCHARGE** pin **[7]**
- Connect the other end to the breadboard's right side power rail **(+)**



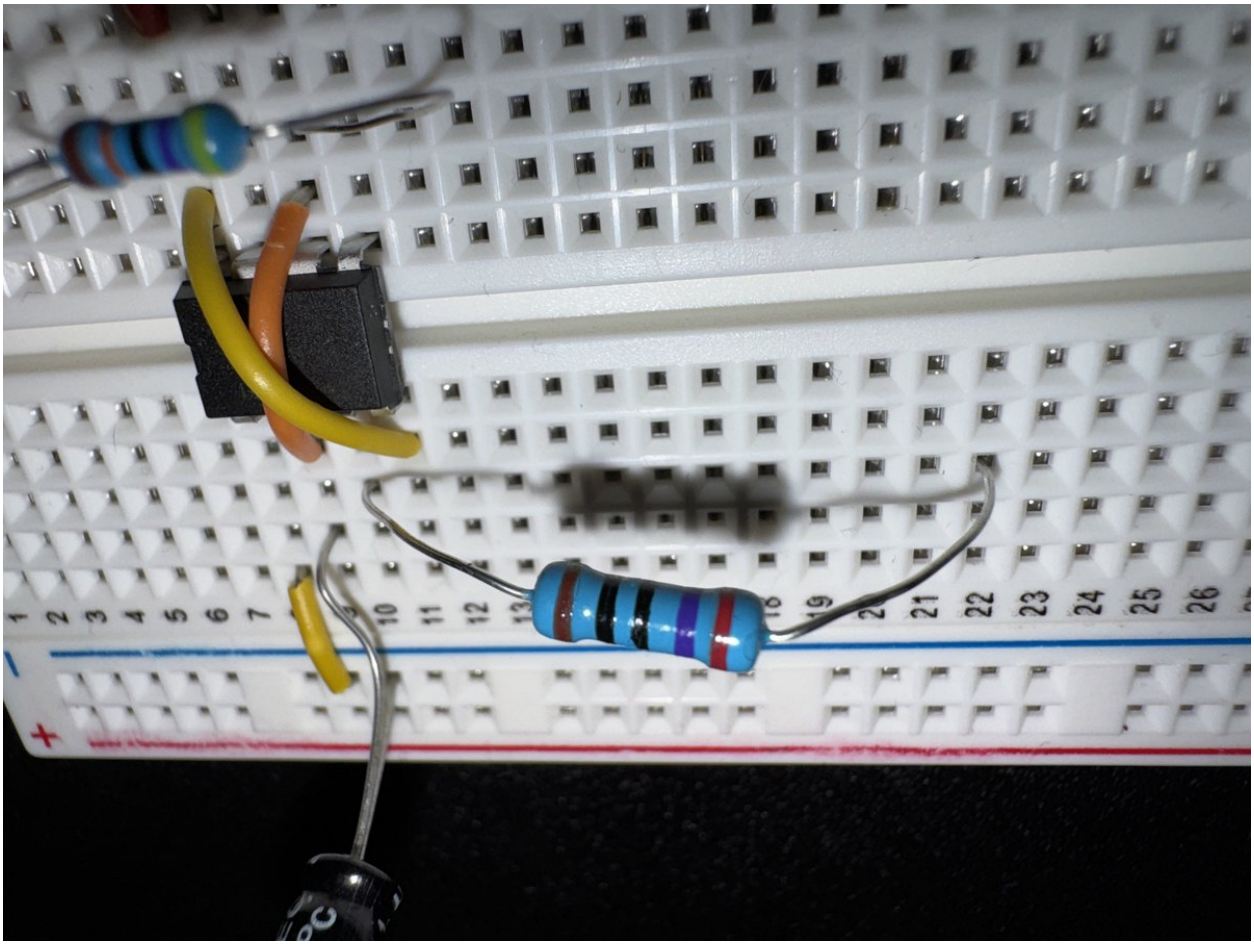
Step 9: Connect 470K Ohm Resistor to 555's DISCHARGE [7] and THRESHOLD [6] pins

- Connect one end of the 470K ohm resistor to the 555's **DISCHARGE [7]** pin
- Connect the other end to the 555's **THRESHOLD [6]** pin



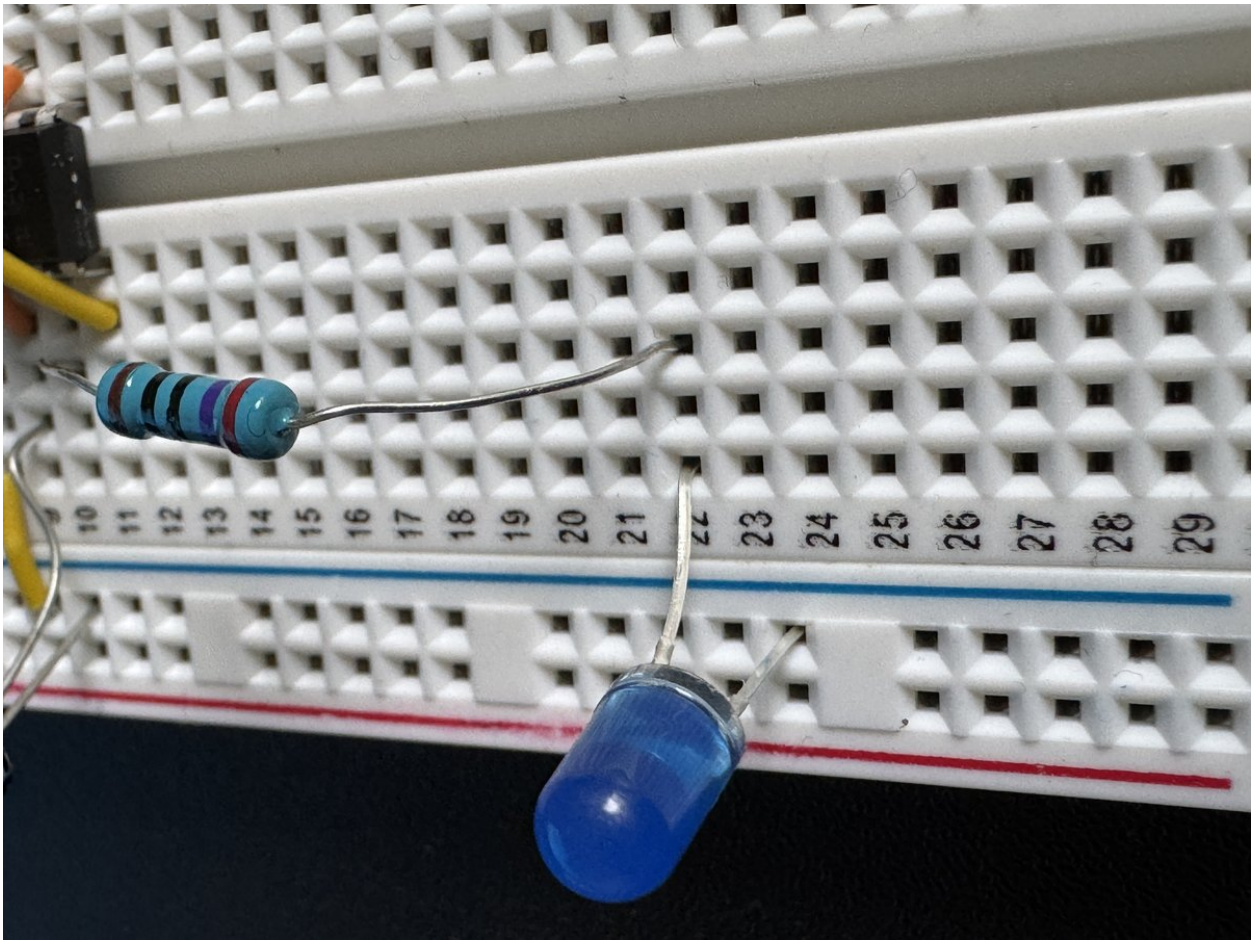
Step 10: Connect the first 270 Ohm Resistor (1 of 2) to 555's OUTPUT [3]

- Connect one end of the **270 ohm resistor** to the 555's **OUTPUT** pin **[3]**
- Connect the other end to any **empty row** that is 5 or more rows below the 555 chip



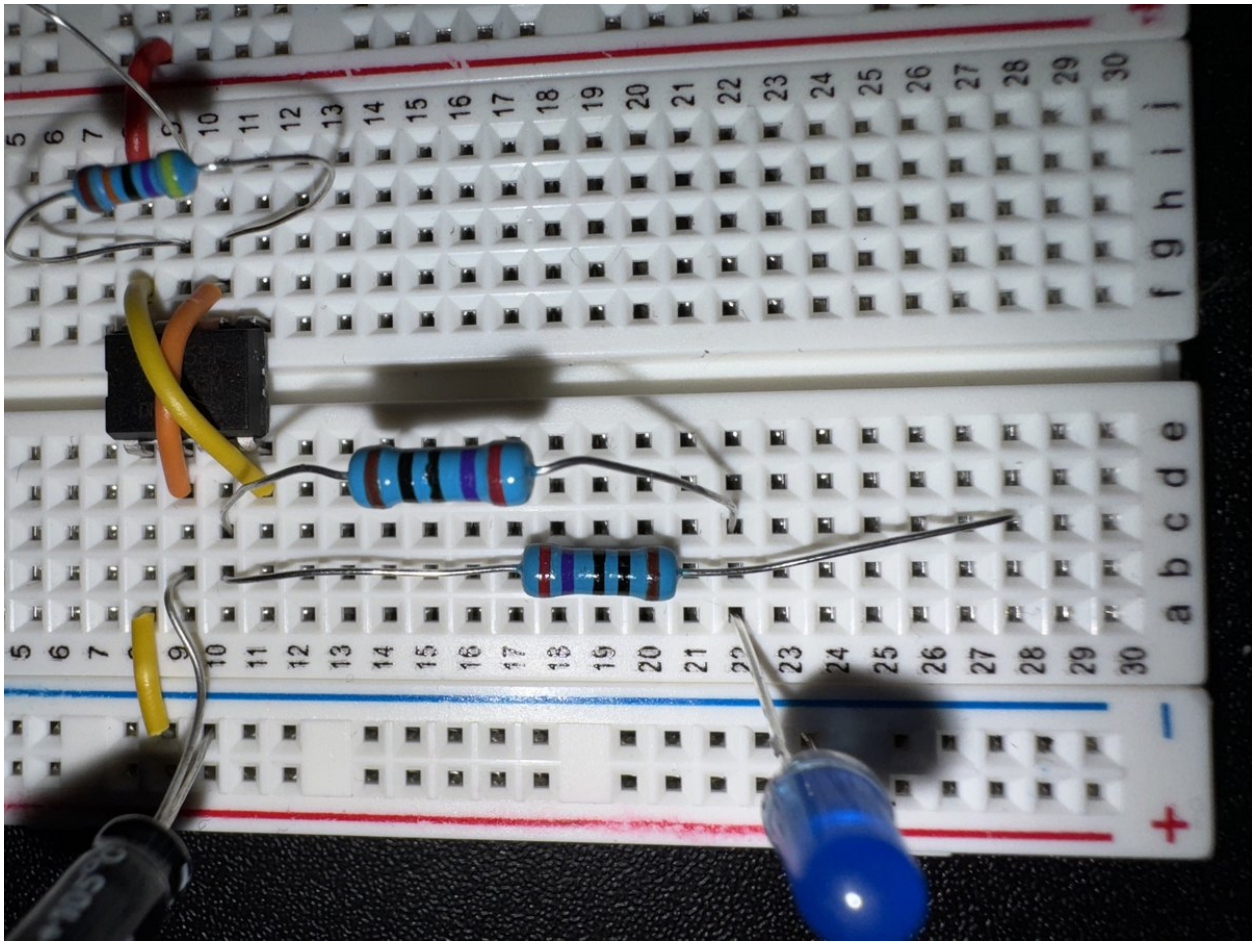
Step 11: Connect LED (1 of 2) to 270 Ohm Resistor #1 and to Ground Rail (-)

- Connect the LED's positive, long leg (anode) to the terminating row of the 270 Ohm resistor from Step 10
- Connect the LED's negative, short leg (cathode) to the breadboard's blue ground rail **(-)**



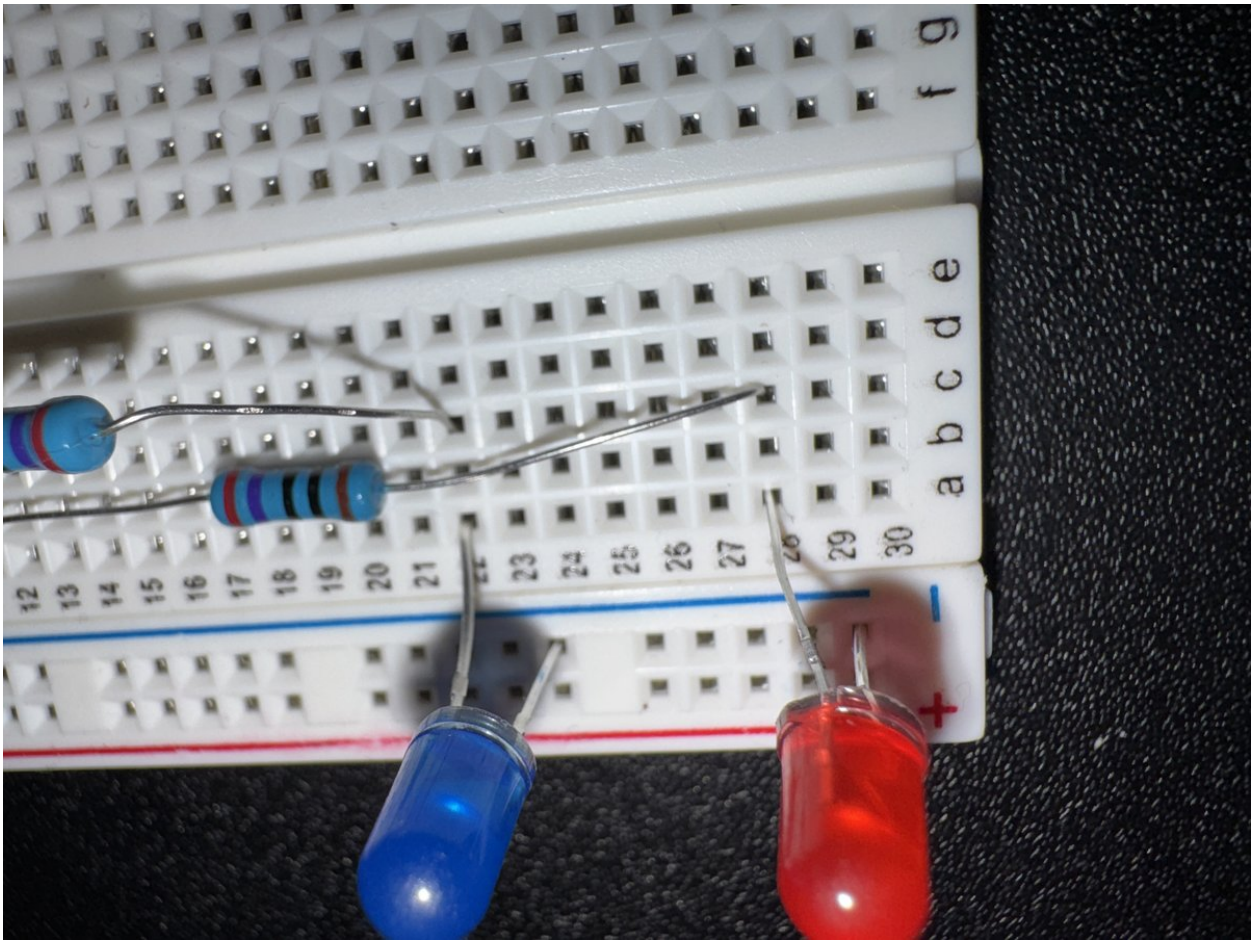
Step 12: Connect the second 270 Ohm Resistor (2 of 2) to 555's OUTPUT [3]

- Connect one end of the **270 ohm resistor** to the 555's **OUTPUT** pin **[3]**
- Connect the other end to any **empty row** that is 5 or more rows below the 555 chip

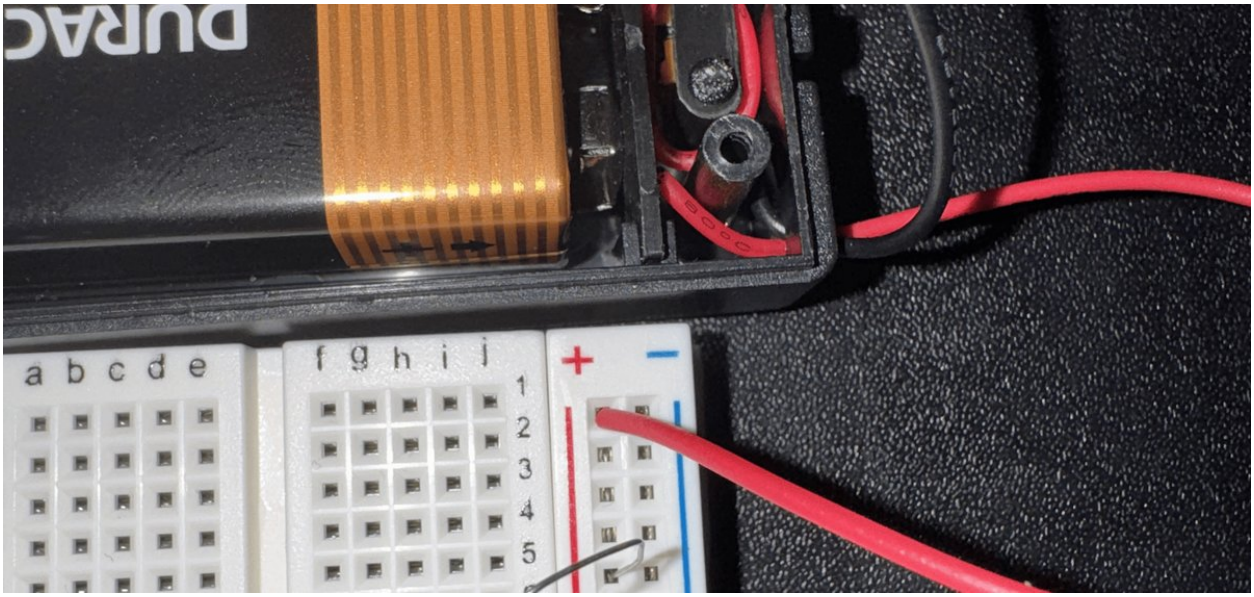


Step 13: Connect LED (2 of 2) to 270 Ohm Resistor #2 and to Ground Rail (-)

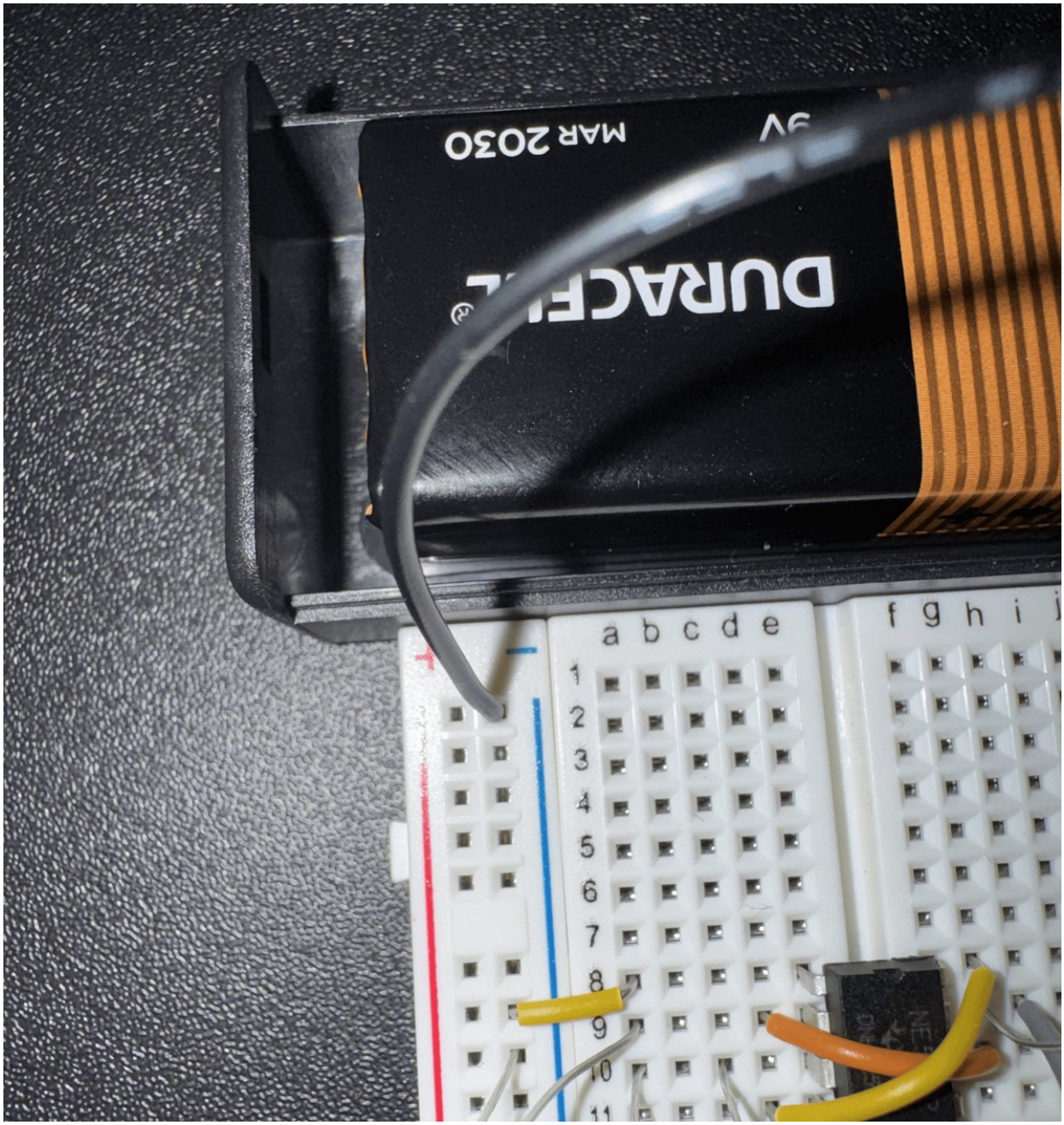
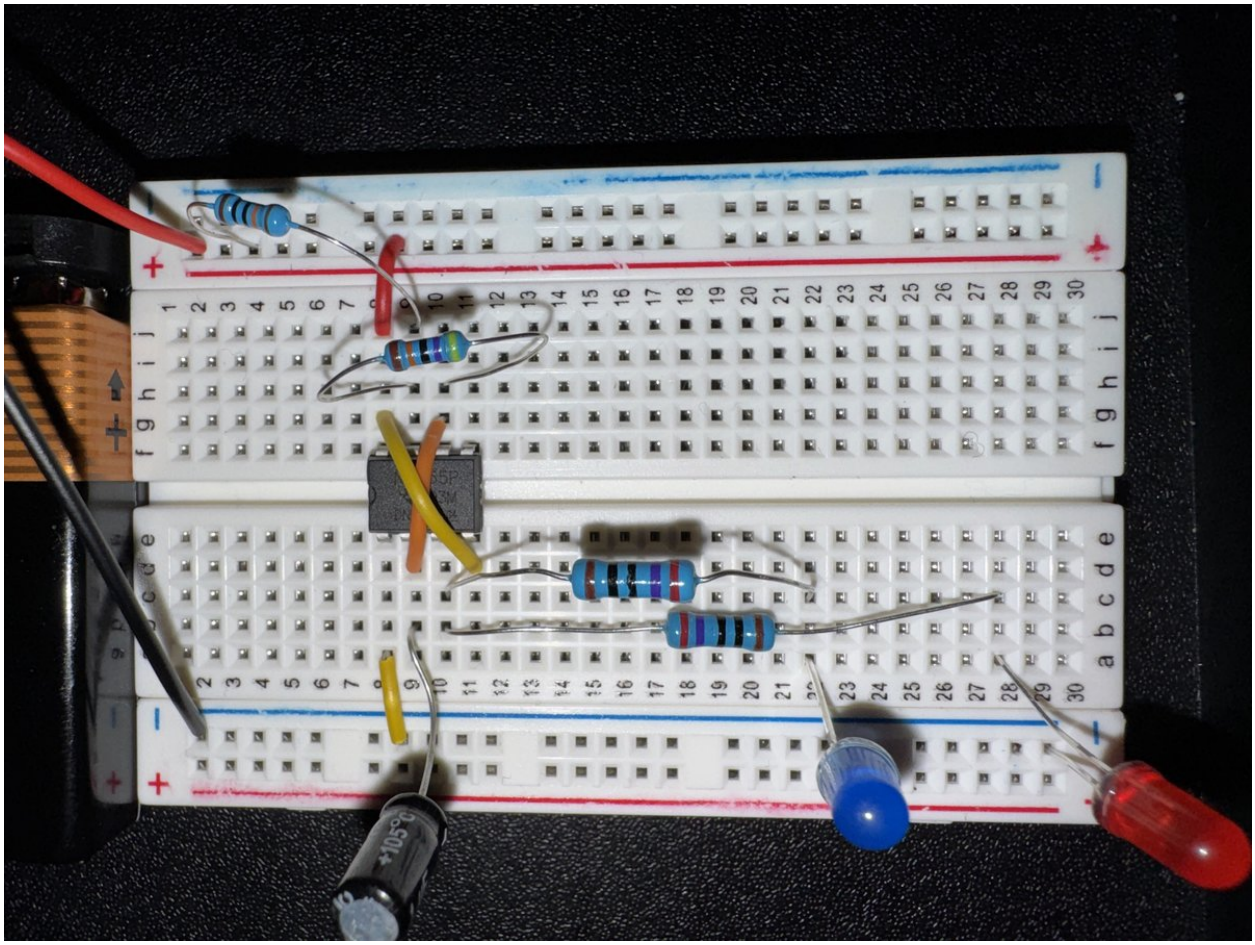
- Connect the LED's positive, long leg (anode) to the terminating row of the 2nd 270 Ohm resistor from Step 12
- Connect the LED's negative, short leg (cathode) to the breadboard's blue ground rail **(-)**



Step 14: Connect 9-Volt Battery's Positive Wire to the Breadboard's Power (+) Rail



Step 15: Connect 9-Volt Battery's Negative Wire to the Breadboard's Ground (-) Rail



The Finished Circuit!